CLASS 10 — MATHS · FORMULA SHEET

Chapter 7: **Coordinate Geometry** | Quick revision, one page

Cartesian plane

- Point (x, y): abscissa x, ordinate y. Quadrants: I $(+,+)$, II $(-,+)$, III $(-,-)$, IV $(+,-)$.
- On the x-axis $y = 0$; on the y-axis $x = 0$.

Distance formula

- $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- Distance from origin: $\sqrt{x^2 + y^2}$.
- Collinear points A, B, C: sum of two of AB, BC, CA equals the third.

Section formula (internal)

- Point dividing $A(x_1, y_1)$, $B(x_2, y_2)$ internally in ratio $m_1 : m_2$:
- $\left(\dfrac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \dfrac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$.

Mid-point formula

- $\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2} \right)$ (section formula with $m_1 : m_2 = 1 : 1$).

Shapes using the distance formula

- Isosceles: two equal sides. Equilateral: three equal sides. Right-angled: $a^2 + b^2 = c^2$.
- Parallelogram: diagonals have the same mid-point. Rectangle: also equal diagonals. Rhombus: all four sides equal. Square: equal sides and equal diagonals.
- Centre of a circle through three points is equidistant from them (set $OA = OB = OC$).

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